

6. Sweet potatoes contain nutrients, including starch, vitamin C and proteins.

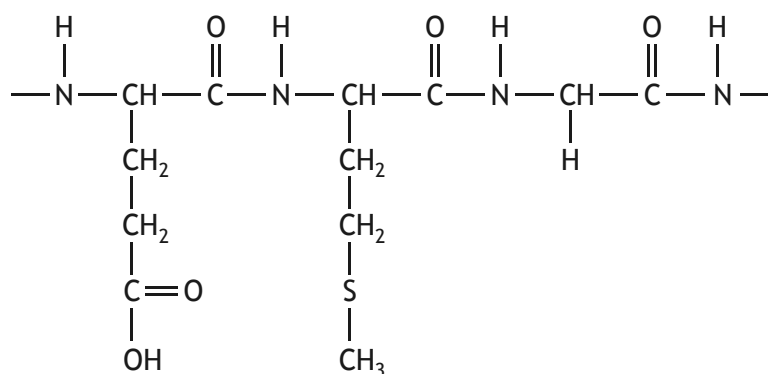
(a) Catalase is an enzyme contained in sweet potatoes that speeds up the breakdown of hydrogen peroxide.

(i) State what is meant by the term enzyme.

1

(ii) Enzymes are a type of protein. Proteins are formed from smaller molecules called amino acids.

(A) A section of a protein is shown.



Circle a peptide link in the above structure.

1

(An additional diagram can be found on *page 33*.)

(B) Draw a structural formula for one of the amino acids used to form this section of protein.

1



\* X 8 1 3 7 6 0 1 1 4 \*

6. (a) (ii) (continued)

(C) State what is meant by the term **essential** amino acid.

1

(D) Name the type of reaction that takes place when amino acids join to form proteins.

1

(iii) As sweet potatoes are cooked, the ability of catalase to break down hydrogen peroxide decreases.

Explain **fully** what happens to the enzyme structure to cause this reduction in activity.

2

[Turn over



6. (a) (continued)

- (iv) Hydrogen peroxide is broken down by catalase to produce water and oxygen.

An experiment was carried out to determine how much oxygen was produced from the breakdown of hydrogen peroxide by a sample of sweet potato.

Draw a diagram showing assembled apparatus that could be used to react hydrogen peroxide solution with sweet potato and measure the volume of oxygen produced.

Your diagram should include labels showing the names and positions of the reactants and the collected product.

3

- (b) Sweet potatoes are a good source of the antioxidant vitamin C.

- (i) Antioxidants like vitamin C are added to food.

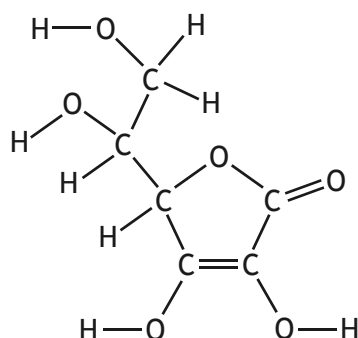
Explain why antioxidants are added to food.

1



6. (b) (continued)

(ii) The structure of vitamin C is shown.



Explain **fully** why vitamin C is soluble in water.

2

- (c) Unlike sweet potatoes, white potatoes contain the chemical solanine, that can be toxic to humans in large doses. A dose of 3 mg per kg of body weight can cause toxic symptoms.

A typical white potato can contain 0.2 mg per g of solanine.

Calculate the mass of white potato that could produce a toxic dose to an adult weighing 65 kg.

3

[Turn over



\* X 8 1 3 7 6 0 1 1 7 \*